Scramble Inversion Beyond Groups: A Ten-Domain Map of Where Denoising Diffusion Solves Sequential Decision Problems

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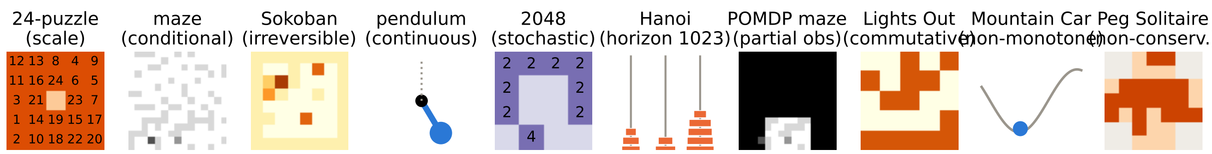

Figure 1: The ten domains, each chosen to remove one property that the Rubik's Cube—where scramble inversion excels [8]—quietly provides. Every panel is a real environment instance from the released code.

Abstract

Scramble inversion—train a denoiser to predict the action that undoes the last step of a random walk from the goal, then act by reverse-process rollout—beats deep approximate value iteration (DAVI) on the Rubik's Cube at a fraction of the compute [8]. Does "every RL problem can be a diffusion problem"? We answer with a ten-domain *assumption ladder*, each domain deleting one property of the cube: group structure (24-puzzle), fixed goal and environment (procedural mazes), invertible actions (Sokoban), discreteness (pendulum), determinism (2048), bounded horizon (Tower of Hanoi, optimal solutions of 1023 moves), full observability (POMDP mazes), ordered solutions (Lights Out), monotone progress (Mountain Car), and conservation (Peg Solitaire). Every domain is run under a matched-architecture, matched-compute protocol against a value baseline on one consumer GPU, with exact oracles wherever the state space permits and mechanical replay verification of every claimed solution. The resulting map is sharp and mechanistic. Scramble inversion *dominates at scale*: 100.0% vs. 0.6% on the 24-puzzle (7.7×10^{24} states) at matched iterations. It survives conditioning, partial observability, commutativity, and non-monotone dynamics. It has two repairable failure modes: with irreversible actions its reverse-only data is deadlock-blind (Sokoban: 65.5% vs. DAVI's 83.1%), which a zero-training hybrid—denoiser proposes, value function vetoes—not only repairs but turns into the best solver (93.0%); and with continuous actions, naive ϵ-regression collapses to a useless posterior mean (0.0% swing-up) while the same data with a distributional head reaches 100.0%. It has one hard boundary: stochastic dynamics (2048), where the deterministic reverse process describes a game that is never played and the denoiser performs at random-play level. We conclude with a practitioner's checklist and release all ten environments, trainers, and logs.

*Experiments, implementation, and manuscript preparation were carried out with the assistance of Claude (Anthropic). Correspondence: `aamer.khani.1@gmail.com`.

#	Domain	Removes	Outcome (greedy, matched)	Verdict
1	24-puzzle	group structure	**denoiser** 100.0% vs 0.6%	✓
2	mazes	fixed goal/env	tie: 76.0% vs 76.6%	✓
3	Sokoban	invertibility	value wins (65.5% vs 83.1%); **hybrid** 93.0%	△
4	pendulum	discreteness	0.0% → 100.0% with distributional head	△
5	2048	determinism	denoiser = random (7.1% vs 9.0%)	✗
6	Hanoi	bounded horizon	2.0%(2.0 – –2.1) vs 1.7%(1.4 – –2.0) (all 59,049 states)	✓
7	POMDP maze	full observability	84.6%(83.3 – –85.3) vs 86.2%(85.9 – –86.9)	✓
8	Lights Out	ordered solutions	100.0%(100.0 – –100.0) vs 33.9%(33.8 – –34.0)	✓
9	Mountain Car	monotone progress	95.8%(92.2 – –99.0) vs 41.8%(0.0 – –66.0)	✓
10	Peg Solitaire	conservation	99.7%(99.5 – –99.8) vs 0.0%(0.0 – –0.0)	△

Table 1: The assumption ladder. ✓ = scramble inversion is doable and competitive or dominant; △ = doable with a specific, identified adaptation (hybrid value veto; distributional head); ✗ = fails for a structural reason. Ranges in the text are min–max over 3 seeds.

1 Introduction

The companion study [8] formalized cube solving as discrete denoising diffusion on a Cayley graph: random moves are the noise, scramble depth is the timestep, and a categorical denoiser trained to invert the last scramble move solves fully scrambled cubes by reverse-process rollout, matching DAVI [1] at 2.8× less wall-clock. That result invites the strong hypothesis, posed by the first author: *every reinforcement-learning problem can be recast as diffusion, and may train faster.* This paper maps the hypothesis's actual extent.

The cube bundles many conveniences at once: a group action, one fixed goal, invertible moves, discrete states and actions, deterministic dynamics, modest solution lengths, full observability, ordered solutions, monotone usefulness of progress, and conservation of the state's "material". Any of these could be the load-bearing one. We therefore built ten environments (Figure 1), each deleting exactly one property, and ran the identical recipe against a matched value-learning baseline in every one. Three principles carried over from the companion paper: (i) *matched protocol*—same residual-MLP backbone, batch, optimizer, and schedule, so the training objective is the only variable; (ii) *exact validation*—five of the ten domains admit exhaustive oracles (8-puzzle: 181,440 states; Hanoi: 59,049; Lights Out: GF(2) algebra; per-instance BFS fields for both maze variants), and every environment ships a test that reproduces a known exact quantity before any training is trusted; (iii) *replay verification*—no solver output is believed until its action sequence is re-executed in the environment.

2 The recipe, its baseline, and one architecture

Recipe. Sample a noising walk from the goal using a reverse transition kernel (the inverse moves themselves when actions are invertible; pulls, un-jumps, un-slides, or reverse-time integration otherwise), with depth $t \sim \mathrm{Unif}\{1..K\}$; train $p_\theta(a \mid s)$ by cross-entropy to predict the forward action that undoes the last noise op; act by greedy rollout (beam search optional). **Baseline.** DAVI: a cost-to-go net trained by one-step bootstrapped targets over the *forward* dynamics, acting by 1-step lookahead—identical backbone, differing only in head and loss. **Architecture** (Figure 2): all ten domains and both objectives share one residual MLP; only the one-hot input width, the head, and the loss change.

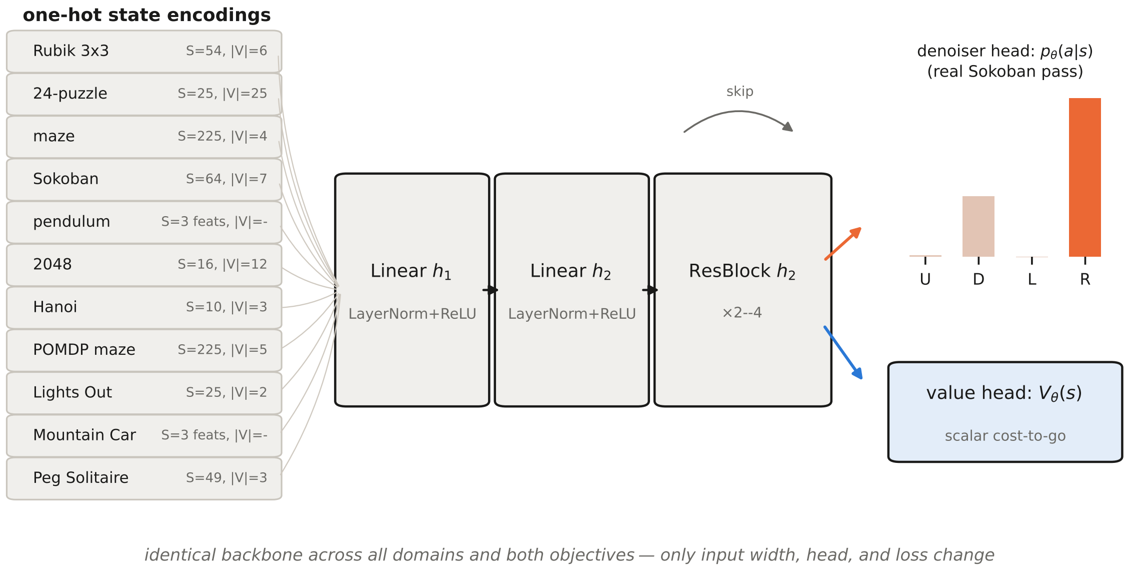

Figure 2: One architecture across the entire study. Left: per-domain one-hot encodings (S tokens, vocabulary $|V|$; the two continuous domains use 3 physical features instead). Middle: the shared LayerNorm residual MLP. Right: the two heads; the denoiser head shows a *real* forward pass of the trained Sokoban policy.

3 Watching noising and denoising in every domain

Figures 3 and 4 are the study in pictures. For each domain the top strip runs the *forward schedule*: the goal state at $t{=}0$ progressively destroyed by the domain's noise operator—random slides, reverse-time torques, pulls, un-jumps, presses—through every intermediate state to full noise at $t{=}1$. The bottom strip runs the *trained reverse process* on a fresh fully-noised instance: each panel is an actual intermediate state of the greedy rollout, ending in the goal (✓) or not (✗). The 2048 row makes the stochasticity failure visible: forward play under random spawns has no relation to the deterministic reverse process the denoiser was trained on, and the board fills with unmergeable debris.

4 Results along the ladder

4.1 Scale is the denoiser's regime (24-puzzle, 8-puzzle, Hanoi)

On the 24-puzzle the denoiser reaches 100% greedy on every probe depth by 60k iterations and solves 500/500 thousand-move scrambles (99.8% with no search); DAVI at matched iterations solves 0.6% (Figure 5): bootstrapped values simply never propagate 100+ moves out in a 7.7×10^{24}-state space. The control puzzles calibrate the claim: on the exhaustible 8-puzzle both methods solve 100% of all 181,440 states (3-seed greedy: denoiser 100.0%, DAVI 100.0%) and DAVI is nearly move-optimal (99.5% of its moves reduce true distance vs. 88.7%)—in small spaces value learning is the sharper tool. On Hanoi, whose 59,049 states we also sweep exhaustively but whose solutions run to 1023 moves, the denoiser attains 2.0% (2.0–2.1) versus DAVI's 1.7% (1.4–2.0): extreme horizon with a tiny state space behaves like the 8-puzzle, not the 24-puzzle—it is *state-space scale*, not solution length, that separates the methods.

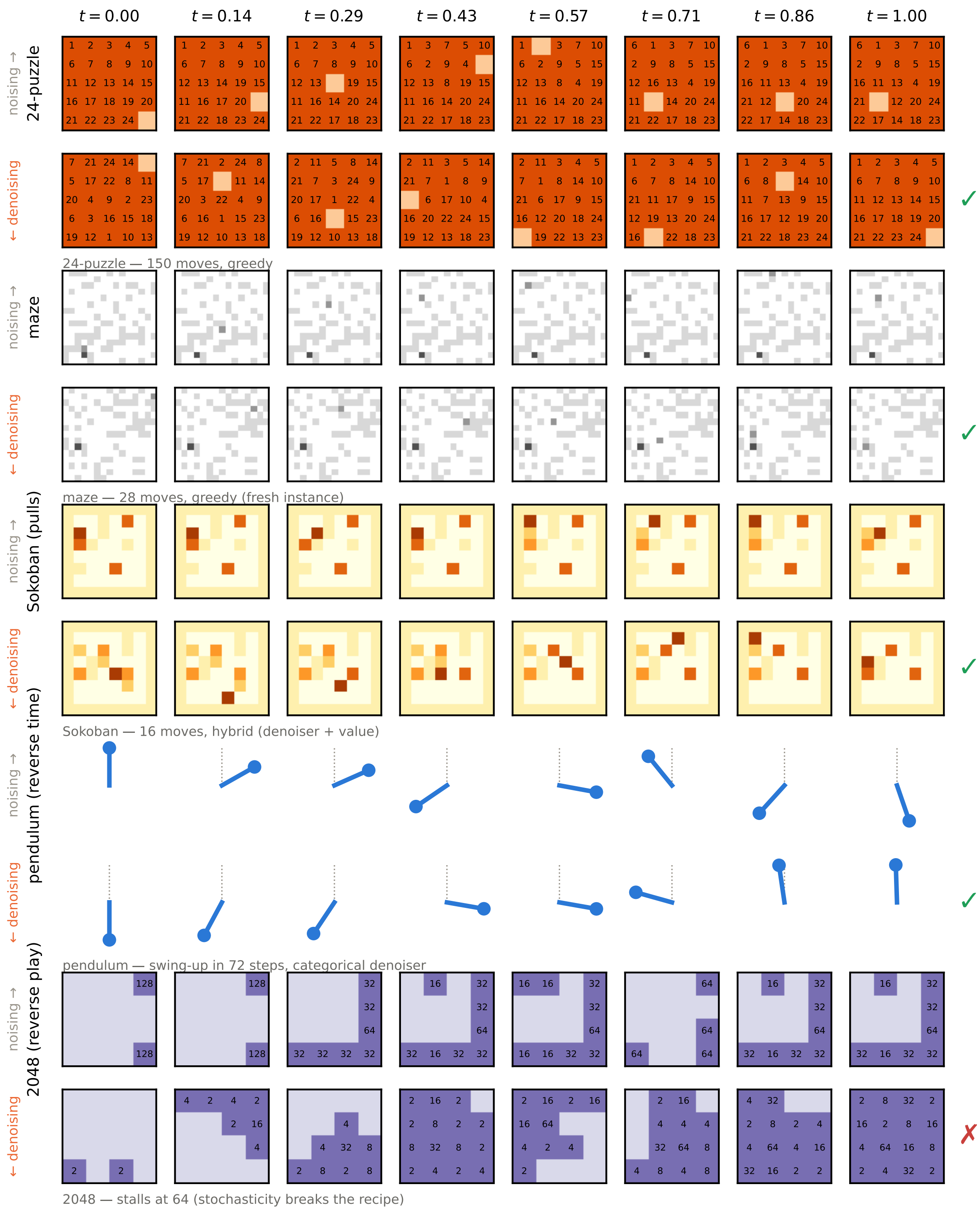

Figure 3: Noising and denoising, domains 1–5. Top strip of each pair: the forward (noising) schedule from the goal, with intermediate states at the labeled t. Bottom strip: the trained denoiser's actual greedy trajectory from a fresh fully-noised state back to the goal (replay-verified; Sokoban row uses the hybrid). The 2048 denoising row fails by construction of the domain, not of the run.

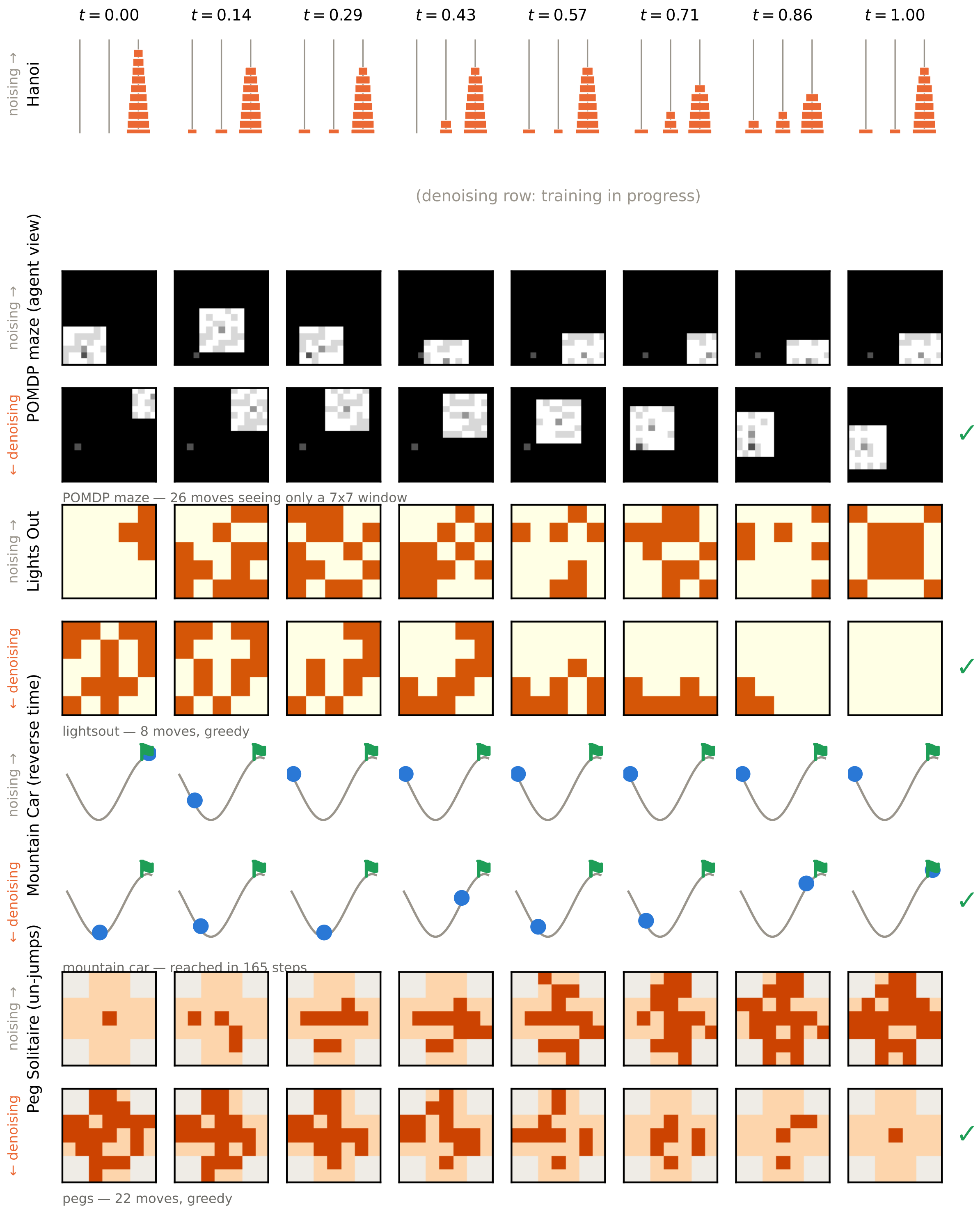

Figure 4: Noising and denoising, domains 6–10: Hanoi, POMDP maze (the strip shows the agent's masked view), Lights Out, Mountain Car (reverse-time noising visible as the car being carried away from the flag), and Peg Solitaire (noising = un-jumps that add pegs; denoising = jumps that remove them).

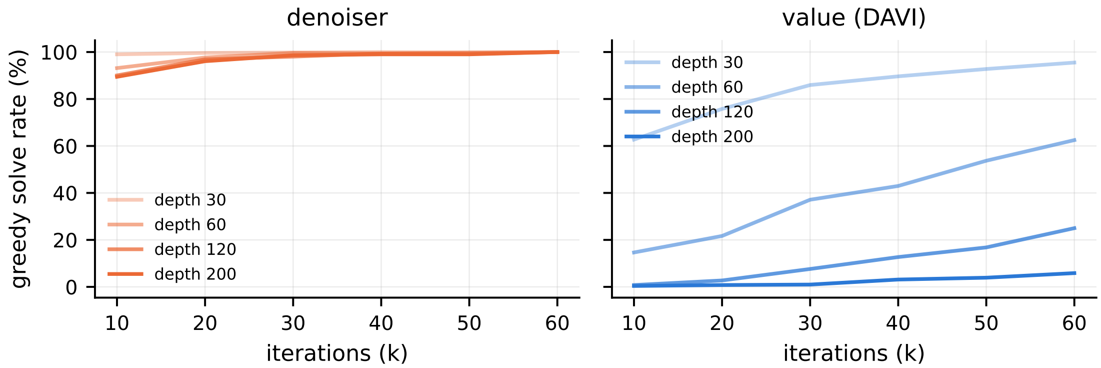

Figure 5: 24-puzzle greedy solve rate during training, by scramble depth (left: denoiser; right: DAVI, same axes and budget).

4.2 Conditioning and partial observability are not boundaries

Procedural mazes (fresh walls and goal each instance; exact per-instance BFS oracles) end in a statistical tie: 76.0% vs. 76.6% on 20k held-out instances, with DAVI's paths closer to optimal (1.05× vs 1.21×) and the denoiser 4× faster to evaluate. Restricting the policy to a 7×7 window (POMDP) degrades both methods together (84.6% (83.3–85.3) vs 86.2% (85.9–86.9)): the information bottleneck, not the objective, is binding. Conditional denoising works as conditional diffusion suggests it should.

4.3 Irreversibility: the failure with a constructive fix (Sokoban, Peg Solitaire)

Reverse (pull-based) walks cannot enter deadlocks, so the denoiser's data contains no signal about fatal pushes: it loses to DAVI at every depth (65.5% vs 83.1% at pull-depth 60, Figure 6). The fix requires *no new training*: act by $\arg\max_a[\log p_\theta(a \mid s) - \lambda V(s'_a)]$—the denoiser proposes a direction, the value net (which bootstraps through forward dynamics and therefore knows dead states) vetoes it. With λ chosen on a held-out split, the hybrid solves 93.0% at depth 60, *surpassing both parents*. Peg Solitaire, whose jumps destroy a peg each move (nothing is conserved and deadlocks abound), replays the pattern at rung ten: denoiser alone 99.7% (99.5–99.8), DAVI 0.0% (0.0–0.0) (depth-25 instances).

4.4 Continuity and non-monotonicity (pendulum, Mountain Car)

Naive *ϵ-regression* on the undoing torque fails absolutely (0.0% swing-up): near hanging, reverse walks spin both ways, the posterior over undoing torques is bimodal, and its MSE-optimal mean is zero torque. The identical data and backbone with a 21-bin categorical head reaches 100.0% (value baseline: 100.0%). The requirement is distributional output—exactly why diffusion models sample rather than regress—not discreteness. Mountain Car then confirms that *non-monotone* progress is no obstacle: reverse-time noising naturally demonstrates the momentum-building detour, and the categorical denoiser reaches the flag in 95.8% (92.2–99.0) of episodes (value baseline 41.8% (0.0–66.0)).

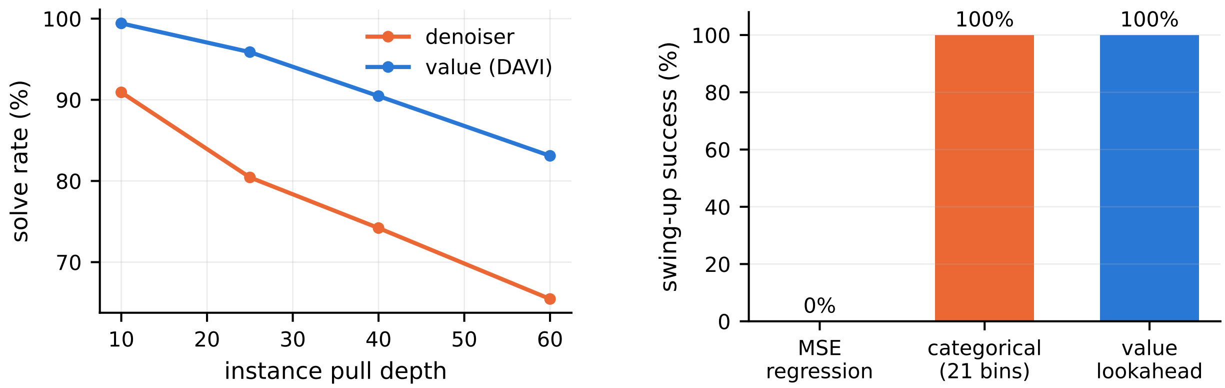

Figure 6: **Left:** Sokoban solve rate vs. difficulty; reverse-only training is deadlock-blind. **Right:** the pendulum ablation — ϵ-regression collapses to the bimodal posterior's mean; the same data with a 21-bin categorical head restores 100%.

4.5 Commutativity (Lights Out)

When solutions are unordered sets, "the move that undoes the last noise op" is maximally ambiguous—any member of a minimal solution set is correct. The Bayes-optimal denoiser simply spreads mass over that set, and greedy rollout peels it off one press at a time: 100.0% (100.0–100.0) solve rate (DAVI 33.9% (33.8–34.0)), with solution lengths checked against the exact GF(2) optimum. Label ambiguity is averaged over, not compounded.

4.6 Stochasticity: the hard boundary (2048)

Trained on deterministic reverse play, the denoiser scores at random-play level in the real game (7.1% vs. random's 9.0% reaching the 256 tile), while a one-line greedy-merge heuristic reaches 61.8%. The mechanism is structural: with exogenous randomness, the reverse process's marginal corresponds to no realizable policy's state distribution, and its undo-labels answer questions the forward game never asks. Repairing this demands a fundamentally different reverse process (marginalizing over spawn randomness), not a tweak.

5 Discussion: a practitioner's checklist

The strong hypothesis fails; a precise and more useful statement survives. Before reaching for scramble inversion, ask:

1. **Can you run the dynamics backward from the goal, deterministically?** If randomness is exogenous (2048), stop; nothing short of a stochastic-aware reverse process will do.
2. **Is the state space too large to bootstrap a value function across?** If yes (cube, 24-puzzle), scramble inversion is likely the *best* tool, not merely a viable one. If the space is small (8-puzzle, Hanoi), value learning will match it and beat it on optimality.
3. **Are actions irreversible?** Expect deadlock-blindness; budget for the value-veto hybrid, which in our experiments is strictly the best policy (Sokoban: 93.0%).

4. **Are actions continuous?** Keep the distribution: discretize or sample; never regress the mean.
5. Conditioning, partial observability, unordered solutions, extreme horizons, and non-monotone progress required *no* adaptation in our experiments.

Limitations. Three seeds per configuration (single seed for the 24-puzzle, where the gap is 166×); one hardware platform; MLP backbones only; greedy/1-step solvers at evaluation (stronger search lifts both methods); Sokoban and Peg Solitaire instances from one generator family; 2048 baselines are heuristic rather than learned.

6 Reproducibility

Ten vectorized pure-PyTorch environments, each with a validation gate that reproduces a known exact quantity (state counts, diameters, GF(2) null-space dimension, reverse-step exactness, replay identities). Machine-readable results in `paper2_data/`; every trainer, evaluator, and figure script at https://github.com/aamirkhani/rubiks-diffusion. Total compute: ∼35 GPU-hours on one RTX 5080 laptop.

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A Hyperparameters

Domain	S	$\|V\|$	M	h_1/h_2/blocks	batch	iters	K
24-puzzle	25	25	4	5000/1000/4	10k	62k	300
8-puzzle	9	9	4	1024/512/2	10k	20k	31
maze / POMDP	225	4 / 5	4	2048/1024/3	8192	60k	60
Sokoban	64	7	4	2048/1024/3	8192	80k	40
pendulum	3 feats	—	21 bins	256×3 SiLU	8192	30k	200
2048	16	12	4	2048/1024/3	8192	40k	40
Hanoi	10	3	6	1024/512/2	8192	20k	1200
Lights Out	25	2	25	2048/1024/3	8192	20k	25
Mountain Car	3 feats	—	3	256×3 SiLU	8192	20k	250
Peg Solitaire	49	3	196	2048/1024/3	4096	30k	31

Table 2: Per-domain configuration. Optimizer Adam (10^{-3}, $\times 0.1$ at 80%), bf16 autocast throughout; DAVI target-net sync at loss < 0.05 or staleness cap. Value baselines share every setting.

B Validation gates

Each environment must reproduce a known exact quantity before training: 8-puzzle BFS = 181,440 states, diameter 31; Hanoi BFS = 3^{10} states, goal eccentricity $2^{10}-1 = 1023$; Lights Out GF(2) null space of dimension 2; maze BFS field = Manhattan distance on empty grids; Sokoban and Peg Solitaire reverse walks replay to the goal under forward dynamics; pendulum and Mountain Car reverse steps invert forward steps to $<10^{-6}$; 2048 slides conserve tile mass; 24-puzzle scrambles preserve permutation parity by construction.